

\* Calculate sum of all elements in given range  $[L, R]$ :-

Overs: 1 2 3 4 5 6 7 8 9 10

Scores: 2 8 14 29 31 49 65 79 88 97



Cumulative Sum

Total runs scored in overs  $i^{\text{th}}$  to  $j^{\text{th}}$   
 $\text{score}[j] - \text{score}[i-1]$

Idea:- Store the cumulative sum of your array.

$$\text{pf}[i] = \text{sum}[0 \dots i]$$

\* Construct prefix sum:-

$$\text{pf}[i] = \text{pf}[i-1] + A[i] \quad , i > 0$$

$$\text{TC} : O(N+Q)$$

$$\text{SC} : O(N)$$

### \* Equilibrium Index:-

$A = a_0 \ a_1 \ a_2 \ \dots \ a_{i-1} \ a_i \ a_{i+1} \ \dots \ a_{N-2} \ a_{N-1}$   
 left sum right sum  
 $\text{sum}[0 \ i-1]$   $\text{sum}[i+1 \ N-1]$

$$\text{leftSum} = \text{pf}[i-1]$$

$$\text{rightSum} = \text{pf}[N-1] - \text{pf}[i]$$

Tc: OCN)

Sc: 0 (N)

\* 0-1 prefix sum:-

$a[10] = 2 \quad 4 \quad 3 \quad 7 \quad 9 \quad 8 \quad 6 \quad 5 \quad 4 \quad 9$

| q | r | ans |
|---|---|-----|
| 0 | 4 | 2   |
| 4 | 8 | 3   |

$$T_C: \mathcal{O}C(\mathbb{Q} + \mathbb{N})$$

$S_C: O(N)$